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Machine learning for neural & biological systems — rigorous measurement of representations, from cortical populations to single-cell atlases.

NeurIPS 2026 · Sydney, 6–13 Dec — NeurReps poster #46, Fri 11 Dec, ICC Sydney Hall 4 · Open to research roles & collaborations in AI × life science

I work on AI for Science — using AI to do science, and doing the science needed to trust what it says. As AI Director at SCMP, I lead AI strategy and implementation, putting generative AI into daily editorial production. Before that, I brought AI and AIoT to local industry at FLAIR; led the team deploying CNN models inside a robotic colonoscope that reached clinical-trial stage at NISI; and co-founded Haalthy, an AI platform for lung-cancer patients. Earlier: R&D at Nokia and Microsoft. MPhil in Information Engineering, CUHK.

My research starts from one discipline: representations — in brains, cells and machines — should be measured before they are believed. I work on measurement guards for subspace claims in neural data; geometric and topological machine learning for trajectories and dynamics, from protein folding to cyclone paths; higher-order interactions in cancer co-mutation, beyond pairwise prediction; and a C. elegans virtual-cell programme — single-cell atlases and perturbation analysis — alongside work on small-RNA inheritance. The conviction running through it is the one behind AI for Science: the coming decade's science belongs to people who can frame the question, build the model and design the analysis with AI in the loop — and who keep measurement as the guardrail that keeps the loop honest.

1 · NEURREPS 2026 — EXTENDED ABSTRACT #46 (THIS POSTER)

Three guards for subspace claims: chance, stability and metric

Question. Do evoked neural population patterns live inside the spontaneous-activity subspace? Classics said yes; modern large-scale recordings say near-orthogonal. The field's standard "containment" number is not one well-posed quantity.

Three guards. Report excess over an explicit rotation null — a random k-dim subspace already captures k/n of any probe (a 10–30-point floor at classic scales). Screen conditioning by the boundary gap σ_k/σ_{k+1} — median gaps fall below 1.05 beyond k 3–5, while published work uses k = 10–128. Declare the metric — rate normalization moves hippocampal cross-state excess from 0.32–0.61 to 0.02–0.19.

Result. The literature's disagreement decomposes into chance-floor inflation, ill-conditioned projectors and undeclared metric + state mixing; the small V1 excess survives all guards. Data: 60 Neuropixels populations, ~10k-neuron two-photon V1, monkey DMFC, HD thalamus, hippocampal CA1; measures ship with a 15-test known-answer battery.

2 · RECOMB 2027 — SUBMITTED (EASYCHAIR #22)

Separating higher-order information from prediction in cancer co-mutation

Most "higher-order" claims in biology rest on pairwise statistics. This work asks how much of cancer co-mutation structure is genuinely beyond pairs.

It separates two quantities that are often conflated: the higher-order information present in the data versus the predictive gain a model can actually extract from it.

Solo-authored; code at github.com/freemanhui/beyond-pairs.

3 · C. ELEGANS — SINGLE-CELL PROGRAMME

A worm virtual cell: atlas, state models & a small-RNA study

C. elegans is the only animal with a complete cell lineage and a complete connectome (302 neurons) — a whole-organism testbed the large labs leave alone.

Atlas direction: a worm-specific embedding and an open cell-type annotation tool on Arc scBaseCount data (295,473 cells, 58 samples), with a daf-2 ageing-perturbation case study under strict held-out validation — baselines first, per the Virtual Cell Challenge lesson.

Companion paper: "The inheritance state reweights the small RNA pool beyond uniform gain in C. elegans" — manuscript in submission at Briefings in Functional Genomics (Oxford University Press).

METHODS TOOLKIT

Representation geometry · information theory · topological data analysis (persistent homology) · single-cell ML